



NURUL KHAUSAR FAQIHAH BINTI MOHD SUHAIMI

Final Year Student

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PROFESSIONAL SUMMARY

Final-year Intelligent Systems Engineering student with hands-on experience in Machine Learning, Generative AI, Agentic AI, and Data Analytics. Skilled in developing AI-powered applications using Python, SQL, databases, and web technologies. Experienced in building LLM-powered applications, AI agents, and workflow automation solutions using Python, APIs, Retrieval-Augmented Generation (RAG), and modern GenAI technologies. Seeking an internship in AI Automation, Generative AI, or AI Engineering. Availability: Available from 7 September 2026 for a 3–6 month internship.

EDUCATION

Bachelor of Information System (Hons.) Intelligent Systems Engineering (Big Data Track) | 2023 – Present

Universiti Teknologi MARA (UiTM) | CGPA: 3.65 | Dean list (semester 1 – semester 5)

Foundation in Engineering | 2022-2023

Pusat Asasi UiTM Dengkil | CGPA: 3.67

FINAL YEAR PROJECT

EduTutor Chatbot For High School Addmath Subject Support Using Large Language Model (LLM) AND Agentic AI

- Developed an end-to-end LLM-powered multi-agent AI tutoring system comprising Router, Tutor, and Hint Agents for personalized SPM Additional Mathematics learning.
- Implemented a Retrieval-Augmented Generation (RAG) system using Qdrant and curriculum-aligned learning materials to improve response relevance and knowledge accuracy.
- Deployed a local AI infrastructure using Docker and Ollama, enabling efficient management and execution of large language models.
- Automated AI workflows using n8n and integrated cloud-based AI services through Ngrok to support external model access and orchestration.
- Developed backend APIs and mathematical reasoning tools to generate solution explanations, formula calculations, and adaptive hints for students.
- Integrated PostgreSQL-based conversational memory to maintain learning context and support personalized tutoring interactions.
- Designed and implemented responsive web interfaces using HTML, CSS, and JavaScript to enhance user learning experience.

SELECTED PROJECTS

Student Stream Placement System | Web-based Decision Support System | 2025

- Developed a fuzzy logic-based decision support system to recommend suitable academic streams using IMK and JKEP assessment data, improving consistency in student placement decisions.
- Performed data preprocessing, fuzzy rule evaluation, and recommendation analysis to generate accurate placement outcomes, visualized through an interactive dashboard built with the Anvil Framework.

Eczema Care Expert System | Web-based Intelligent System | 2025

- Developed a rule-based expert system that analyzes skin characteristics and symptoms to recommend personalized skincare guidance.
- Implemented inference rules and database integration to automate consultations and improve recommendation consistency.

SVR Optimization Using Particle Swarm Optimization (PSO) | 2025

- Optimized Support Vector Regression (SVR) using Particle Swarm Optimization (PSO) for automated hyperparameter tuning.
- Evaluated prediction performance using optimized model parameters to improve forecasting accuracy on energy consumption datasets.

Sentiment Analysis on Learning Language Application | 2025

- Developed a sentiment analysis model using Natural Language Processing (NLP) techniques to classify user feedback from language learning applications.
- Analyzed sentiment classification results to identify key user satisfaction factors and generate actionable insights for application improvement.

Telco Customer Churn Classification Model | 2025

- Built a machine learning classification model to predict customer churn using demographic and behavioral data.
- Performed exploratory data analysis (EDA) and identified customer retention patterns to support predictive analytics and business decision-making.

SKILLS

Programming Language: C, C++, Java, Python, R, SQL

AI and Machine Learning: Weka, RapidMiner, Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG)

AI Automation, Workflow Orchestration & Deployment: n8n, Docker, Ollama, Azure AI Foundry, Ngrok

Web and Application Development: HTML, CSS, JavaScript, Anvil Framework

Databases: MySQL, PostgreSQL, Qdrant

Backend & APIs: REST API Development, API Integration, JSON

UI/UX Design: Figma

Tools: Visual Studio Code, Git, GitHub

EXTRA-CURRICULAR ACTIVITIES

Vice Treasurer | SULAM Community Program, Universiti Teknologi MARA | 2026

- Managed program budgeting and expenditure tracking to support AI awareness and business digitalization workshops for local entrepreneurs.
- Coordinated logistics and collaborated with committee members to ensure successful community engagement activities.

AJK 3K | UiTM Brass Band | 2025

- Managed venue readiness, equipment organization, and safety arrangements during band rehearsals and events.
- Collaborated with committee members to support the smooth execution of performances and university activities.

Bagpipe Section Member | UiTM Brass Band | 2023 - 2025

- Participated in rehearsals, performances, and official university events, including PERKAD and LIBK.
- Strengthened teamwork, discipline, and time management through over two years of ensemble participation.

RECOGNITION AND AWARDS

CREATIONS DE UiTM International Mega Innovation Carnival |2023

- Developed an RFID-based accessible parking management system for OKU users.
- Awarded Silver Medal for innovation and practical system implementation

Database programming with SQL-Oracle Academy |2024

- Completed Oracle Academy Database Programming with SQL certification.
- Achieved successful completion of the final assessment.

REFERENCE

Farah Syazwani binti Mohamed Rashid | Lecturer | Universiti Teknologi MARA (UiTM) | Faculty of Computer and Mathematical Sciences | [LinkedIn](#) | farahsyazwani@gmail.com